

MASA: Multimodal Federated Learning through Modality-Aware and Secure Aggregation

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Abstract—As a promising paradigm, federated learning has been applied to multimodal sensing tasks due to its deployment convenience. However, the recent advances in multimodal federated learning emphasize learning a high-quality multimodal model but overlook the model usage requirements of massive unimodal clients. Moreover, the privacy risk in model sharing and client data heterogeneity impact the efficacy of federated learning. In this paper, we propose a novel multimodal federated learning system named MASA. As a departure from existing approaches, MASA simultaneously enhances the model learning efficiency of both multimodal and unimodal clients while ensuring their data privacy. Firstly, we employ a gated cross-modal distillation scheme to achieve performance-aware knowledge transfer across modality-heterogeneous clients. To enhance the system security, MASA integrates a lightweight split-shuffle mechanism to realize the anonymization and encryption of model aggregation. Moreover, to reach personalized collaboration while protecting privacy, MASA features an attention-based spontaneous client clustering mechanism to form client cluster structures securely and distributedly. We evaluate our MASA on four public multimodal datasets for human activity recognition. The results show that our MASA outperforms leading multimodal federated learning methods on the model performance of both multimodal and unimodal clients.

Index Terms—Multimodal sensing, federated learning, knowledge distillation.

1 INTRODUCTION

Multimodal sensing approaches¹ play a crucial role in enhancing the efficiency and accuracy of sensing systems across various domains, such as health monitoring [25], autonomous driving [45], and human-computer interaction [8]. These approaches are characterized by their ability to integrate diverse sensor types, thereby merging multiple data streams to improve overall sensing performance. Prior works have focused on training a “one-fit-for-all” multimodal model for boosting the performance of sensing system [7]. Such centralized approaches require a labor-intensive and time-consuming process of collecting labeled training samples. In addition, the original labeled data may contain private user information, and its centralized processing could give rise to significant privacy concerns [25]. Consequently, to date, few large-scale multimodal datasets that involve humans have been collected and made public.

Federated learning (FL) [37] has recently attracted attention as a distributed machine learning paradigm, which can

mitigate labor and time costs by aggregating the information collected from distributed devices [18]. Significantly, FL decreases privacy leakage since it only uploads model parameters (rather than original data) from the local model to the server [38]. State-of-the-art works [14], [25], [36], [41], [44] aim to deliver a high-quality model by addressing modality heterogeneity within a multimodal FL system. While important, most of them overlook to emphasize the sensing performance of the clients containing unimodal data. This drawback limits the scale at which the multimodal sensing system can be operated. The issue stems from the fact that unimodal clients typically represent the commercially deployed lightweight IoT devices, widespread in real-world environments (approximately 30 billion devices [29]). These IoT devices typically capture a limited array of data types or even only one data category due to cost constraints, form factor limitations, and computational boundaries. Examples include cameras, smart speakers, and mmWave devices, all of which substantially inhibit the potential benefits of leveraging multimodal systems for sensing performance [25].

In this paper, we ask the following question: Can we push the development of multimodal sensing systems by improving the performance of multimodal and unimodal clients simultaneously without compromising user privacy? We propose MASA, which realizes the vision by designing an FL framework to facilitate the aggregation, transmission, and exchange of valuable knowledge collected from distributed clients to enhance the collaboration between clients, as shown in Figure 1. MASA emphasizes enhancing the performance across various types of clients, ensuring scalability to dynamic changes within the system.

To achieve the proposed vision of MASA, overcoming the modal heterogeneity across various clients is impera-

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1. A multimodal sensing system seamlessly integrates diverse sources and types of data, including multimodal and unimodal sensors.

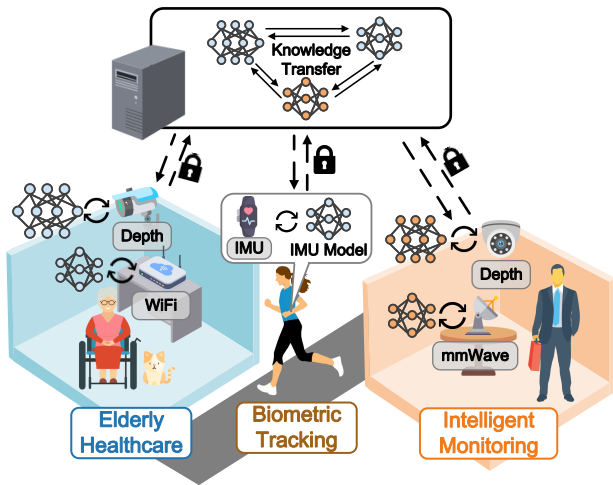


Figure 1: MASA designs a federated learning framework to facilitate the knowledge transfer between unimodal and multimodal clients, which contributes to various downstream applications.

tive. Therefore, we design a gated cross-modal distillation (GCMD) scheme, enabling multimodal clients to select the most superior modality by evaluating the representation performance. To preserve the feature space of each modality, we define a sample correlation matrix, serving as the medium of distillation. In addition, to avoid the superior modality negatively affected by other inferior ones, MASA uses a gate unit to control the distillation direction, ensuring the superior knowledge flows into inferior models unilaterally.

Traditional FL preserves all raw signal data on the distributed devices and only uploads the model parameters to the server, reducing privacy leakage risk. Unfortunately, despite these measures, MASA faces a vulnerability due to the plaintext transmission of model parameters, potentially allowing for the straightforward recovery of the original data [24]. To tackle this challenge, we introduce a simple but effective solution, namely a split-shuffle mechanism (SSM), to ensure malicious servers cannot decode the model parameters and recover the original data even if they can intercept the transmission. With SSM, MASA can employ information encryption while the client uploads the model parameters to the server. The key idea behind SSM involves establishing a middle node between clients and the server to ensure client anonymization. SSM dynamically splits each client model into two meaningless random components, which are then uploaded to the server. By doing so, SSM can intentionally transform the transmission of local model parameters from plaintext to ciphertext. Therefore, MASA improves the performance and reliability of the system. It is worth noting that low-computation SSM can be deployed on various devices, such as smartphones, wireless routers, more innovative gateways, or lightweight IoT devices.

While advantageous for enhancing privacy, the utilization of ciphertext transmission introduces a limitation by preventing the FL framework from exchanging and learning knowledge using the server to boost the sensing performance. The restriction arises because the server can only

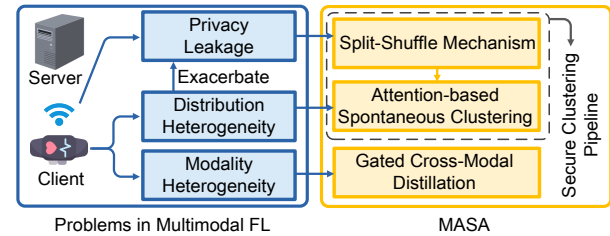


Figure 2: Challenges in multimodal federated learning and corresponding solutions integrated in MASA.

cluster and aggregate the plaintext parameter uploaded from clients according to their data distributions, and then send the most analogous cluster model back to the clients. This process aims to mitigate the well-known model bias issue and achieve multi-model assistance [3]. To address this issue, we propose a novel solution named the attention-based spontaneous client clustering (ASCC) scheme. Our key insight is delivering the similarity-matching cluster selection from the server to the client. In MASA, the server exclusively recombines and aggregates random components into cluster models, and sends them to each client for updating its local model. The local model, leveraging a few data, calculates the fitness and autonomously selects the valuable model parameters to improve its performance. The resulting solution not only enhances the sensing performance of unimodal clients but also benefits multimodal clients. This is achieved by the sharing and fluidity of knowledge among different clients, thereby creating a collaborative enhancement across the system.

We take the human activity recognition task as an example to build a MASA prototype and evaluate it on four commonly used public datasets, including 71 users, 56 activities, and 6 types of sensors. Experimental results demonstrate that MASA surpasses leading multimodal FL methods, achieving an increase in average accuracy ranging from 8.89% to 11.16%. We further show that MASA is robust, delivering consistently good performance in challenging situations where clients exhibit diverse types. Figure 2 illustrates the components integrated in MASA.

Our main contributions are summaries as follows:

- We propose a multimodal sensing FL framework called MASA that simultaneously enhances the performance of both multimodal and unimodal clients, while bolstering the preservation of user privacy.
- We introduce a local cross-modal distillation scheme, enabling multimodal clients to fully utilize the knowledge of superior modality to boost inferior ones. Through model aggregation, the superior knowledge can also be further transferred into unimodal users with an inferior modality.
- We design a privacy-enhanced spontaneous client clustering mechanism, which constructs client cluster structures in a distributed manner without disclosing client model plaintexts.

2 RELATED WORK

Our proposed MASA aims to jointly consider the heterogeneity manifested in both modality and data distribution, and this section introduces some related advances, including multimodal federated learning and personalized federated learning.

2.1 Multimodal Federated Learning

In the evolving landscape of data modality diversity, numerous FL frameworks have emerged to address the challenges of multimodal data learning. Several studies, for instance, operate under the premise that each client possesses solely unimodal data, with modalities differing across clients [40], [42]. A notable framework proposed by Zhang et al. [42] focuses on unimodal training for multimodal prediction, employing strategies like posterior-based attention to integrating different modalities. Yang et al. [40] further delve into cross-client disentanglement to uncover the interplay between various modalities. Simultaneously, research streams [4], [14], [25], [44], [45] parallel our proposed MASA considering clients with aligned multimodal data. Approaches such as FedMSplit [4] interpret these clients as dynamic graphs, optimizing model aggregation, while AutoFed [45] innovatively reconstructs missing modalities in autonomous driving scenarios. Similarly, FedMEKT [14] and CreamFL [41] implement knowledge distillation and contrastive learning, respectively, for cross-modal knowledge transfer. Nevertheless, their methodologies presume the availability of an unbiased public multimodal dataset among clients, a scenario often impractical for some privacy-sensitive tasks with diverse modality configurations. Other frameworks have explored modality fusion learning for multimodal clients and pursued modality-aware aggregation [25], [44]. However, as demonstrated by Harmony [25], this strategy can lead to model inconsistencies between multimodal and unimodal clients, prompting the application of separated training and late fusion to refine multimodal models. Significantly, the performance of unimodal models tends to be neglected during the fusion process. In contrast, MASA effectively harnesses the capabilities of multimodal clients to simultaneously enhance both multimodal and unimodal client performance, thereby offering a more holistic improvement in system capabilities.

2.2 Personalized Federated Learning

Data heterogeneity among clients in FL systems results in varied local training objectives, negatively impacting overall system performance. To address this challenge, various model personalization strategies have been proposed in FL [21], [23], [34], [35]. These strategies range from layer decoupling, which helps retain local knowledge for clients [21], to model interpolation techniques that aim to strike a balance between generalization and personalization [34]. Another line of studies leverages attention mechanism to search optimal aggregation weights for local models. For instance, Assadi et al. [2] construct a self-attention matrix to capture the correlation of client data distribution. Park et al. [28] specifically design a multi-scale attention module to improve the federated temporal

prediction tasks. Furthermore, methods such as transfer learning [35] and multi-task learning [23] have been adopted to enhance personalization.

For clients exhibiting similar data distributions, clustering strategies are frequently utilized, grouping them into clusters for facilitating local model aggregation [3], [5], [9], [12], [16], [26], [27], [31], [32]. Sattler et al. [32] and Briggs et al. [3] propose iterative clustering based on model cosine similarity, constructing binary tree structures of client similarity from top-to-bottom and bottom-to-top, respectively. In contrast, Ghosh et al. [9] and FedSim [27] use a one-shot K-Means approach [12] for clustering. FlexCFL [5] introduces a novel measurement for model divergence using Euclidean distance of Decomposed Cosine similarity (EDC) to simplify clustering complexity. ClusterFL [26] captures the intrinsic clustering relationships among clients, excluding weakly correlated clients to expedite convergence. Furthermore, advanced soft clustering strategies like FedSoft [31] and FLSC [16] allow clients to have varying membership degrees across different clusters, addressing complex and realistic mixed data distributions and facilitating knowledge transfer between clusters, unlike hard clustering [31]. Nevertheless, these similarity-based clustering strategies often necessitate revealing model plaintexts and client cluster labels, which contradicts the principle of model privacy fundamental to FL. To tackle this, our MASA integrates a spontaneous clustering strategy, which securely establishes a cluster structure using randomly encrypted client models, thereby preserving model privacy.

3 PRELIMINARY

This section first introduces multimodal Federated system model, privacy attack model in federated system, and then provides the problem statement. The detailed symbol definitions are shown in Table 1.

3.1 Multimodal Federated System Model

3.1.1 Heterogeneous Client Data

In our framework, we establish a collection of clients, each denoted as n_i within the set $\mathcal{N} = \{n_i | i \in [1, N]\}$, where N signifies the total number of clients. The system's modalities are encapsulated by $\mathcal{M} = \{m | m \in [1, M]\}$, where M reflects the overall count of modalities considered. For any given client n_i , the modalities accessible to them are specified by a subset $\mathcal{M}_i \subseteq \mathcal{M}$, with the actual number of accessible modalities being $M_i = |\mathcal{M}_i|$. This arrangement underscores the variability in modal access across clients, indicating a diverse distribution of modalities where M_i is not uniform but varies from client to client. This diversity extends to data as well, with each client n_i managing M_i distinct datasets corresponding to their modalities. These datasets, denoted as $\mathcal{D}_i^m$ for modality m , are characterized by aligned samples within each client's modality subset. This setup introduces a layer of complexity, acknowledging that clients might possess overlapping yet distinct models, leading to heterogeneous and unevenly distributed data across the network. Such heterogeneity is further complicated by the fact that even among clients sharing the same set of modalities—a condition known as modality isomorphism—significant disparities in data distribution can arise,

Table 1: Symbol definition.

Symbol	Definition
$\mathcal{N}$	Client group
$\mathcal{M}$	Total modality set
$\mathcal{M}_i$	Modality set available to client n_i
$\mathcal{D}_i^{m,u}$	Unlabeled dataset of n_i for modality m
$\mathcal{D}_i^{m,l}$	Labeled dataset of n_i for modality m
$(x_{i,j}^m, y_{i,j}^m)$	The j -th data-label pair of n_i
$w_{i,t}^m$	Local model for modality m of n_i at the t -th round
$w_g^{t,m}$	Global model for modality m at the t -th round
C	Total class number
L_{rec}	Reconstruction loss function
L_{cls}	Classification loss function
θ_i	Aggregation weight of n_i
T	Maximum training round
$f_{i,j}^m$	Low-dimension feature of $x_{i,j}^m$
$\hat{\mathcal{Y}}_i^{m,1}$	Predicted label set corresponding to $\mathcal{D}_i^{m,1}$
$\mathcal{Y}_i^{m,1}$	True label set corresponding to $\mathcal{D}_i^{m,1}$
H_i^m	Entropy of $\mathcal{Y}_i^{m,1}$
$\tilde{H}_i^m$	Entropy of $\hat{\mathcal{Y}}_i^{m,1}$
ϕ_i^m	Mutual information of $\hat{\mathcal{Y}}_i^{m,1}$ and $\mathcal{Y}_i^{m,1}$
s_i^m	Mutual information score for modality m of n_i
m^*	Superior modality
B_i^m	Sample correlation matrix for modality m of n_i
K	Client cluster number
L_m	Model layer number for modality m
$\mathcal{Q}_i^{m,0}$	Query dataset sampled from $\mathcal{D}_i^{m,1}$
$A_i^{m,l}$	Attention parameters of the l -th layer in K cluster models

as indicated by the condition $p(\mathcal{D}_i^m) \neq p(\mathcal{D}_j^m)$ for any two clients i and j in $\mathcal{N}$. This highlights the unique challenges presented by the diverse and uneven distribution of data within our framework.

Moreover, in general IoT scenarios, devices are deployed for data sensing and processing but cannot automatically annotate the collected data, thus frequently encountering a lack of labeled data. Therefore, we organize the dataset associated with modality m for each client n_i , denoted as $\mathcal{D}_i^m$, into unlabeled and labeled subsets. These are expressed as $\mathcal{D}_i^{m,u} = \{x_{i,j}^m | j \in [1, D_i^{m,u}]\}$ for unlabeled data and $\mathcal{D}_i^{m,l} = \{(x_{i,j}^m, y_{i,j}^m) | j \in [1, D_i^{m,l}]\}$ for labeled data, where $x_{i,j}^m$ represents the features of the j -th sample, and $y_{i,j}^m$, which belongs to $\{1, \dots, C\}$, is its label, where C denotes the total number of classes. $D_i^{m,u}$ and $D_i^{m,l}$ respectively denote the total number of samples in the unlabeled and labeled subsets. Given the significantly smaller size of $D_i^{m,l}$ compared to $D_i^{m,u}$.

3.1.2 Multimodal Federated Learning

This subsection introduces the workflow of multimodal FL, which is an iterative process of local training and model aggregation.

Model Initialization. First, the server defines initializes the global models $\mathcal{W}_g^0 = \{w_g^{m,0} = (e_g^{m,0}, d_g^{m,0}, c_g^{m,0}) | m \in \mathcal{M}\}$ for total M modalities, where $e_g^{m,0}$, $d_g^{m,0}$ and $c_g^{m,0}$ are the initialized encoder, decoder and classifier for the modality m , respectively. Then the server broadcasts $w_g^{m,0}$ to client n_i if $m \in \mathcal{M}_i$.

Local Training. To fully leverage the potential of abundant unlabeled data, semi-supervised learning approaches are applied to the local training stage. For each client n_i and modality m within $\mathcal{M}_i$, an auto-encoder is trained using

the unlabeled dataset $\mathcal{D}_i^{m,u}$. This auto-encoder consists of an encoder $e_i^{m,t}$ and a decoder $d_i^{m,t}$, where t denotes the local training round.

During the unsupervised training phase, the optimization objective can be expressed as:

$$e_i^{m,t}, d_i^{m,t} = \arg \min_{e_i^{m,t-1}, d_i^{m,t-1}} \sum_{j=1}^{D_i^{m,u}} L_{rec} \left(d_i^{m,t-1} \left(e_i^{m,t-1} (x_{i,j}^m) \right), x_{i,j}^m \right). \quad (1)$$

The reconstruction loss L_{rec} , such as Mean Squared Error (MSE) loss, aims to extract general but informative representations. Once trained, $e_i^{m,t}$ employs $\mathcal{D}_i^{m,l}$ to update the local classifier c_i^m and further fine-tune its parameters, as detailed in Eq. (2) with an emphasis on cross-entropy loss L_{cls} for classification:

$$c_i^{m,t} = \arg \min_{c_i^{m,t-1}} \sum_{j=1}^{D_i^{m,l}} L_{cls} \left(c_i^{m,t-1} \left(e_i^{m,t} (x_{i,j}^m) \right), y_{i,j}^m \right). \quad (2)$$

Notably, for multimodal clients, models corresponding to different modalities are trained independently. We refrain from using modality fusion training [13], due to its proven incompatibility in heterogeneous modality settings within FL [25].

Model Aggregation. After local training, clients upload their local models to the server. The server aggregates models from the same modality, where $\mathcal{N}_m$ denotes the clients possessing modality m by:

$$w_g^{m,t} = \left\{ \sum_{n_i \in \mathcal{N}_m} \theta_i e_i^{m,t}, \sum_{n_i \in \mathcal{N}_m} \theta_i d_i^{m,t}, \sum_{n_i \in \mathcal{N}_m} \theta_i c_i^{m,t} \right\}. \quad (3)$$

The aggregation weight for each client n_i , denoted θ_i , is proportional to their volume of training data. The sum of the aggregation weights $\sum_{n_i \in \mathcal{N}_m} \theta_i = 1$.

Model Broadcast. Following aggregation, the server disseminates the updated global models $\mathcal{W}_g^t = \{w_g^{m,t} | m \in \mathcal{M}\}$ back to the clients, marking the beginning of the next round of local training. This cycle repeats until the system round t reaches the predetermined maximum round T .

3.2 Privacy Attack Model

In this paper, we mainly consider privacy attacks especially sample reconstruction attack launched by the curious server. This subsection provides the corresponding attack model, including the goal and the ability of attacker, the basic process of sample reconstruction attack and the indicators to measure the privacy leakage.

Attacker Goal. The goal of attacker (i.e., curious server) is to reconstruct the private training samples by analyzing the model parameters uploaded by clients. Please note that: the server is honest-but-curious in our considered scenarios [15], which only tries to reveal client privacy but will not interfere with the federate training. This differs from poison attacks.

Attacker Ability. The cloud server is able to communicate with any client and obtain its local model. Moreover, as an insider of FL, the server also knows the configurations of federated training, e.g., training sample size, data class number, loss function type and model architecture [47]. We assume the server masters the classical sample reconstruction

techniques, such as Deep Gradient Leakage (DGL) [47], [43] and infers the privacy of individual clients based on the received model parameters in the model aggregation stage. Specifically, in DGL, the server randomly initializes a batch of trainable tensors as the samples to be reconstructed. These random samples are input to the received model, and the dummy gradient is generated from the loss. By reducing the distance between the true gradient (i.e., the difference of the global model and the received model) and the dummy gradient, the random samples are updated and gradually closer to the true samples [47].

Attack Indicators. To evaluate the privacy attacking effect, we apply two indicators in this paper: Privacy Leakage Amount (PLA) and Sample Reconstruction Error (SRE) [20]. PLA refers to the amount of parameters in any client model which are successfully recovered by the server. SRE is the mean squared error of the reconstructed samples and the corresponding true samples. Higher PLA and higher SRE both represent better attack effect.

3.3 Problem Statement

The traditional multimodal FL system models are trained and aggregated separately for different modalities. These approaches fail to achieve effective knowledge transfer between modalities because they can not extract separable features from the aligned multimodal data on multimodal clients. The shortfall is further compounded by the variable abilities of different modalities to extract separable features. Additionally, the conventional practice of aggregating isomorphic local models into a singular global model fails to measure the biases introduced by pattern heterogeneity among clients, leading to a one-size-fits-all global model that may not align with each client's unique data distribution, hindering improvements in sensing performance. Last but not least, traditional FL methods use plaintext transmission of model parameters, which can potentially lead to privacy leakage.

4 SYSTEM DESIGN

This section presents the overview and the detailed design of MASA, including the gated cross-modal distillation, split-shuffle mechanism, and attention-based spontaneous client clustering.

4.1 System Overview

As illustrated in Figure 3, MASA integrates three essential designs to concurrently improve the performance of both multimodal and unimodal clients while enhancing privacy protection. These designs include the Gated Cross-Modality Distillation (GCMD), the Split-Shuffle Mechanism (SSM), and the Attention-based Spontaneous Client Clustering (ASCC). GCMD proves especially advantageous for clients with multiple modalities, facilitating the enhancement of inferior modalities by leveraging knowledge from superior ones. For privacy maintenance, each client's local model is transferred to a reliable shuffler, where it undergoes anonymization and segmentation using SSM prior to its aggregation on the cloud server. Upon the distribution of aggregated models to clients, the ASCC approach is

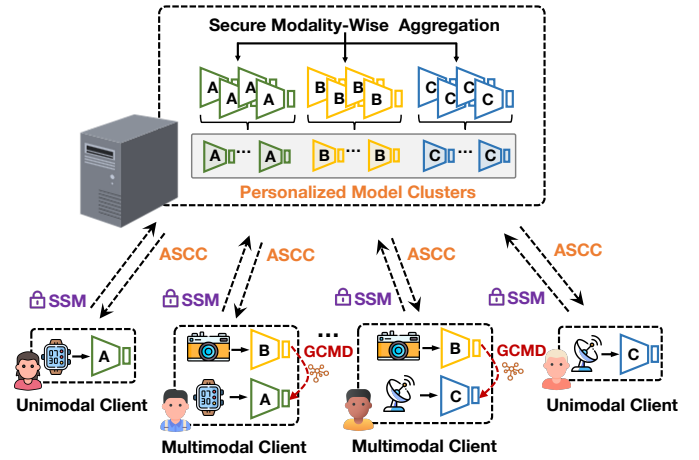


Figure 3: Architecture of MASA.

employed to reevaluate and select the cluster model that optimally aligns with their local data distribution. This cyclical process persists until the client models converge, ensuring a balanced improvement in both performance and privacy.

4.2 Gated Cross-Modal Distillation

The volume and range of information vary significantly across modalities, which may lead to differences in the convergence speeds and accuracies of models trained on distinct modalities. This is because different data modalities offer diverse perspectives on sensing tasks. For instance, in the context of human activity recognition, image or skeleton data comprehensively capture the positions of all joints, whereas smartwatch sensors only specifically track wrist movement trajectories. However, in many cases, it is hard to accurately sense user activity solely based on the motion trajectory of a single joint. Therefore, compared with sensor data, skeleton data reflects the characteristics of human activities more comprehensively, and generally achieves higher sensing accuracy.

To validate this phenomenon, we conduct a preliminary experiment using the MHAD dataset [6] which comprises both IMU and skeleton modalities. We randomly select a client from this dataset and train multimodal model with independent IMU encoder and skeleton encoder. By T-SNE dimension reduction [22], The features extracted by these two encoders are depicted in Figure 4 and Figure 5. From these visualizations, it is observable that, in comparison to the IMU model, the skeleton model demonstrates enhanced clustering performance for samples within the same class. The model accuracy, assessed using the two distinct modality encoders and depicted in Figure 6, shows the skeleton model converges much faster than the IMU model even on the same client.

This finding encourages us to exploit the differences in representational performance between modalities to facilitate cross-modal knowledge transfer. Our key idea is to leverage the knowledge from superior modalities as an auxiliary supervisory signal to facilitate the convergence of

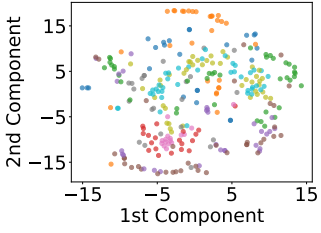


Figure 4: Visualization of IMU features.

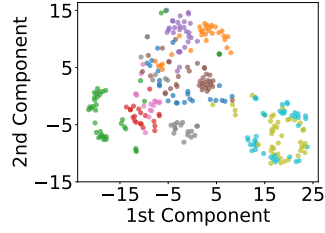


Figure 5: Visualization of skeleton features.

models in inferior modalities. To achieve this goal, we need to address the following two critical challenges: i) How to determine the relative superior or inferior of each modality during local training, and ii) How to transfer cross-modal knowledge across highly differentiated feature spaces between modalities. Please note that the superiority of any modality is not absolute or constant, but will dynamically change with the sensing environment and task types.

Therefore, we introduce the gated cross-modal distillation (GCMD) scheme, as depicted in Figure 7, GCMD encompasses two main components: superiority measurement and sample correlation distillation, which are designed to address the identification of superior or inferior modalities and the knowledge distillation across modalities, respectively. The detailed process is presented in Algorithm 1.

To determine the distillation direction and ensure the knowledge flows from superior modalities to inferior ones, we first define a metric named Mutual Information Score (MIS) to measure the superiority of each modality. First, each multimodal client n_i applies the labeled dataset of modality m ($m \in \mathcal{M}_i$) $\mathcal{D}_i^{m,l}$ to the encoder e_i^m , and then n_i gets the feature set $\mathcal{F}_i^{m,l} = \{f_{i,j}^m | j \in [1, D_i^{m,l}]\}$, where $f_{i,j}^m = e_i^m(x_{i,j}^m)$. To evaluate the performance of e_i^m to separate the samples from different data classes, n_i first clusters the feature vectors in $\mathcal{F}_i^{m,l}$ via K-Means [12]², and the cluster labels of feature vectors is denoted as a set $\mathcal{Y}_i^{m,l} = \{\tilde{y}_{i,j}^m | j \in [1, D_i^{m,l}]\}$. Next, n_i computes the mutual information ϕ_i^m between $\mathcal{Y}_i^{m,l}$ and the true label set $\mathcal{Y}_i^{m,l} = \{y_{i,j}^m | j \in [1, D_i^{m,l}]\}$:

$$\phi_i^m = \sum_{c=1}^C \sum_{\tilde{c}=1}^C P_i^{m,\text{jnt}}(c, \tilde{c}) \log_2 \left(\frac{P_i^{m,\text{jnt}}(c, \tilde{c})}{P_i^{m,\text{mar}}(c) P_i^{m,\text{mar}}(\tilde{c})} \right). \quad (4)$$

where C is the total class number in $\mathcal{Y}_i^{m,l}$. $P_i^{m,\text{jnt}}(c, \tilde{c}) = p(c, \tilde{c} | \mathcal{Y}_i^{m,l}, \mathcal{Y}_i^{m,l})$ is the joint probability mass function (PMF), $P_i^{m,\text{mar}}(c) = p(c | \mathcal{Y}_i^{m,l})$ and $P_i^{m,\text{mar}}(\tilde{c}) = p(\tilde{c} | \mathcal{Y}_i^{m,l})$ are the marginal PMFs of $\mathcal{Y}_i^{m,l}$ and $\mathcal{Y}_i^{m,l}$, respectively.

Then, n_i calculates the MIS s_i^m of the modality m :

$$s_i^m = \frac{\phi_i^m}{H_i^m + \tilde{H}_i^m}. \quad (5)$$

where H_i^m and $\tilde{H}_i^m$ is the entropy of the PMFs of $\mathcal{Y}_i^{m,l}$ and $\mathcal{Y}_i^{m,l}$, respectively. H_i^m can be written as:

$$H_i^m = - \sum_{c=1}^C P_i^{m,\text{mar}}(c) \log_2 P_i^{m,\text{mar}}(c). \quad (6)$$

2. Other clustering methods such as Mean-Shift [11] are also applicable.

The MIS s_i^m represents the correlation between $\mathcal{Y}_i^{m,l}$ and $\mathcal{Y}_i^{m,l}$. In other words, higher s_i^m means that more samples from the same class are divided into one cluster, therefore the performance of e_i^m in separating features from different classes is better. Therefore, comparing the MIS s_i^m of all modalities, the distillation gate determines the modality m^* with the highest MIS as the superior modality.

After determining the distillation direction, each multimodal client n_i integrates the cross-modal distillation into unsupervised local training. Inspired by similarity-based distillation [30], in each update of the local encoder e_i^m , n_i samples a batch b_i^m from the unlabeled dataset $\mathcal{D}_i^{m,u}$ in each modality. Please note that for different modalities such as m_1 and m_2 , $b_i^{m_1}$ and $b_i^{m_2}$ are aligned. Besides the common reconstruction loss L_{rec} on b_i^m , the encoders trained on inferior modalities are further regularized by the distillation loss term:

$$L_{\text{dis}} = \|B_i^{m^*} - B_i^m\|. \quad (7)$$

where B_i^m is a Sample Correlation Matrix (SCM) with the size of $|b_i^m| \times |b_i^m|$. Each element $B_{i,(j,k)}^m$ in B_i^m is the inner-product of feature vectors $f_{i,j}^m$ and $f_{i,k}^m$, which implies the correlation between two samples. Due to the performance differences measured above, the SCM output by the superior encoder $e_i^{m^*}$ contains more accurate information about classification. Therefore, we let the rest inferior encoders learn the superior knowledge from $B_i^{m^*}$ to further rectify their encoding patterns. For the superior modality m^* , only L_{rec} is maintained to avoid the negative impact of L_{dis} on its representation performance:

$$L_i^m = \begin{cases} L_{\text{rec}}(e_i^m, d_i^m | \mathcal{D}_i^{m,u}), & \text{if } m = m^* \\ L_{\text{rec}}(e_i^m, d_i^m | \mathcal{D}_i^{m,u}) + L_{\text{dis}}(e_i^m | \mathcal{D}_i^{m,u}), & \text{otherwise} \end{cases} \quad (8)$$

As shown in Figure 8, employing knowledge distillation from superior modalities to inferior modalities can significantly enhance model accuracy. The design of GCMD showcases several distinct features and benefits, emphasizing privacy security, knowledge transferability, and adaptability to multimodal data. Its key strength lies in privacy preservation, operating entirely within multimodal users and avoiding any external exposure of private data or features, thereby allowing for the creation of a local Trusted Execution Environment (TEE) for enhanced security. Unlike other cross-client distillation methods, GCMD is accessible and straightforward to implement, as it does not require a public dataset for knowledge transfer. Significantly, GCMD enhances inferior modalities in multimodal clients, potentially transferring superior knowledge from superior modalities to unimodal clients with inferior ones through model aggregation. Moreover, implementing unsupervised local training and distillation in GCMD, predicated on sample correlation, augments its applicability for knowledge transfer across disparate feature spaces.

4.3 Split-Shuffle Mechanism

Following local training using the GCMD, each client n_i with the modality m sends its local model to the cloud server for modality-aware aggregation. However, this direct plaintext transmission between clients and the server poses significant data privacy risks. If a malicious server intercepts

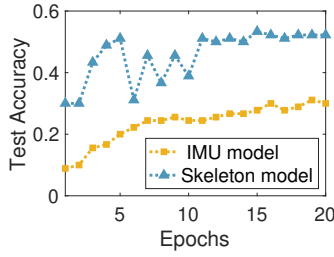


Figure 6: Test accuracy of IMU model and skeleton model.

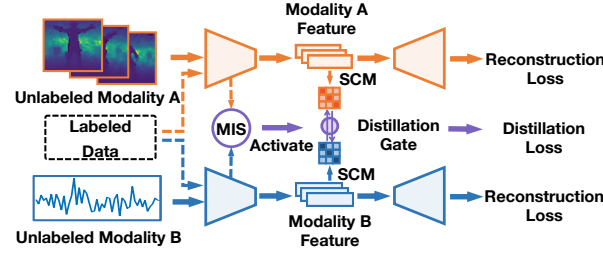


Figure 7: The workflow of gated cross-modal distillation (GCMD) scheme.

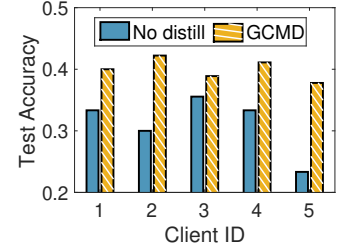


Figure 8: Improvement of local IMU model applying GCMD.

Algorithm 1: Gated Cross-Modal Distillation Scheme

Input: (1) client group $\mathcal{N}$; (2) encoder e_i^m and decoder d_i^m ; (3) unlabeled dataset $\mathcal{D}_i^{m,u}$ and labeled dataset $\mathcal{D}_i^{m,l}$; (4) update rounds T ; (5) distillation loss weight α ; (6) learning rate μ

Output: e_i^m and d_i^m for modality m of n_i

```

1 for  $n_i \in \mathcal{N}$  do
2   //  $n_i$  is a multimodal client
3   if  $M_i > 1$  then
4     /* Superiority Measurement */
5     for  $m \in \mathcal{M}_i$  do
6        $s_i^m \leftarrow \text{computeMIS}(e_i^m, \mathcal{D}_i^{m,l})$ ; // Eq. 5
7       // select the modality  $m^*$  with the highest  $s_i^{m^*}$  as
8       // superior modality
9        $m^* \leftarrow \arg \max_s (s_i^1, \dots, s_i^{M_i})$ ;
10      /* Sample Correlation Distillation */
11      for  $t = 1, \dots, T$  do
12        for  $m \in \mathcal{M}_i$  do
13          Sample a random batch  $b_i^{m,t}$  from  $\mathcal{D}_i^{m,u}$ ;
14          // obtain features of sample batch  $b_i^{m,t}$ 
15           $f_i^{m,t} \leftarrow e_i^m(b_i^{m,t})$ ;
16          // obtain reconstructed sample batch
17           $\hat{b}_i^{m,t} \leftarrow d_i^m(f_i^{m,t})$ ;
18          // compute reconstruction loss
19           $L_{\text{rec}} \leftarrow \text{computeMSE}(\hat{b}_i^{m,t}, b_i^{m,t})$ ;
20          // compute SCM of  $b_i^{m,t}$ 
21           $B_i^{m,t} \leftarrow f_i^{m,t} \cdot f_i^{m,tT}$ ;
22          if  $m \neq m^*$  then
23            // compute distillation loss
24             $L_{\text{dis}} \leftarrow \text{MSE}(B_i^{m,t}, B_i^{m^*,t})$ ;
25             $L_i^{m,t} \leftarrow L_{\text{rec}} + \alpha L_{\text{dis}}$ ; // Eq. 7-8
26          else
27             $L_i^{m,t} \leftarrow L_{\text{rec}}$ ;
28          // update  $e_i^m$  and  $d_i^m$  using gradient descent
29           $e_i^m \leftarrow e_i^m - \mu \nabla L_i^{m,t}(e_i^m)$ ;
30           $d_i^m \leftarrow d_i^m - \mu \nabla L_i^{m,t}(d_i^m)$ ;
31 return  $e_i^m$  and  $d_i^m$  for modality  $m$  of each client  $n_i$ 

```

these client models, private attributes or even training samples could potentially be reconstructed through inversion attacks, as indicated in [10].

To address this concern, we introduce a Split-Shuffle Mechanism (SSM) during the model aggregation phase, illustrated in Figure 9. First, we assume that there are λ relay nodes between clients and the curious server. In order to shield the local model plaintext from the server,

clients utilize these relay nodes to split and shuffle their local model parameters. Specifically, after local training, each client n_i splits its local model for the m -th modality $w_i^m = \{e_i^m, d_i^m, c_i^m\}$ to $S_i \leq \lambda$ random components: $S_i^m = \{s_{i,j}^m | j \in [1, S_i]\}$, where $w_i^m = \sum_{j=1}^{S_i} s_{i,j}^m$. Here w_i^m can be simply regarded as a vector which consists the parameters of local encoder, decoder and classifier. Then, client n_i randomly selects S_i relay nodes from total λ ones, and sends S_i random components with the cluster label to these nodes, respectively. This step is called transmission shuffle. Please note that the cluster label is randomly selected by each client in the initialized system round, and will be updated via our proposed iterative clustering optimization. Each relay node aggregates the collected components to the intermediate results according to the cluster labels, shown as $\chi_{i,j}^m$ in Figure 9, and then forwards the intermediate results to the server. The server further aggregates the intermediate results to the final cluster models.

In the above model uploading process applying SSM, any relay node and the cloud server cannot recover the local model plaintexts of clients. By implementing SSM, we ensure that the server does not gain access to any complete model in plaintext and remains unaware of the ownership of any particular model during the model aggregation. The design of SSM, characterized by its easy implementation, strikes a fine balance between ensuring security and maintaining a simple structure. This balance makes it particularly suitable for deployment on relay nodes with limited computational resources, such as wireless routers and gateways. Additionally, in our SSM, it is not required that all relay nodes be trustworthy. Even if relay nodes collude with the server with a certain probability, our SSM can still protect client privacy to a great extent, which is more in line with the reality. In the following, we will analyze in detail the relationship of privacy security and SSM configurations, such as collusion probability and random component number.

Theorem 1: When the server colludes with $j < S_i$ relay nodes, the server cannot recover the local model of client n_i . Assume the collusion probability of each relay node is ρ , the probability of this absolutely secure case is: $\sum_{j=0}^{S_i-1} C_{\lambda}^j \rho^j (1-\rho)^{\lambda-j}$.

Proof: Each client n_i splits its local model w_i^m to S_i random components $s_{i,1}^m, \dots, s_{i,S_i}^m$, and randomly selects S_i relay nodes for the distributed storage of these components. If the curious server aims to recover w_i^m , it should at least collude with the S_i relay nodes selected by n_i . The relay nodes that collude with the server will provide the random

components collected from n_i to the server. However, this contradicts the premise of $j < S_i$. Therefore, the server cannot obtain all the components from n_i . We model the collusion of relay nodes and the server as a binomial distribution problem. The probability of collusion of j relay nodes from total λ ones and the server is: $C_\lambda^j \rho^j (1 - \rho)^{\lambda-j}$, where C_λ^j denotes the combination number of selecting j relay nodes. From the above analysis, as long as it meets $j < S_i$, the server cannot recover the local model of n_i . Therefore, the probabilities of $j \in [0, S_i - 1]$ are summed as $\sum_{j=0}^{S_i-1} C_\lambda^j \rho^j (1 - \rho)^{\lambda-j}$.

Theorem 2: When the server colludes with $j \geq S_i$ relay nodes, and the collusion probability is ρ , the probability of recovering the local model of client n_i is: $\sum_{j=S_i}^{\lambda} C_\lambda^j \rho^j (1 - \rho)^{\lambda-j}$.

Proof: If the S_i relay nodes randomly selected by the client n_i are a subset of the relay nodes that collude with the server, that is, all the relay nodes selected by n_i are malicious, the random components sent by the client n_i will be leaked to the server by the malicious relay nodes, and the server can directly sum them up to recover the plaintext of the client model. For the sake of simplicity in calculating the probability of *Case A*: all the relay nodes selected by n_i are malicious, we first fix the selection of n_i . Assume that S_i relay nodes have been selected by n_i , which consists total $C_\lambda^{S_i}$ combinations. For any combination, to satisfy *Case A*, we assume that the relay nodes selected by n_i all collude with the server. Then the server also needs to collude with other $j - S_i$ relay nodes. Therefore, under the premise of satisfying *Case A*, there are $C_{\lambda-S_i}^{j-S_i}$ combinations for j malicious nodes. Therefore, the total number of combinations that satisfy *Case A* is $C_\lambda^{S_i} C_{\lambda-S_i}^{j-S_i}$. When the client n_i and the server independently choose relay nodes, the total number of combinations is $C_\lambda^{S_i} C_\lambda^j$. Therefore, when the number of collusive nodes is j , the probability of the model plaintext being recovered is: $C_{\lambda-S_i}^{j-S_i} / C_\lambda^j$. From Theorem 1, the probability of j relay nodes colluding with the server is: $C_\lambda^j \rho^j (1 - \rho)^{\lambda-j}$. Therefore, for the probability of the server recovering the local model of client n_i is: $\sum_{j=S_i}^{\lambda} C_\lambda^j \rho^j (1 - \rho)^{\lambda-j}$.

Theorem 3: When total N clients participate in the model uploading applying SSM, and the server colludes with $j \in [0, \lambda]$ relay nodes, the expectation of Privacy Leakage Amount (PLA) is: $\sum_{i=1}^N \sum_{j=S_i}^{\lambda} C_\lambda^j \rho^j (1 - \rho)^{\lambda-j} |w_i^m|$.

Proof: According Theorem 1 and 2, for each client n_i : (a) if $j < S_i$, the model privacy will not be disclosed; (b) if $j \geq S_i$, the probability of model leakage is: $\sum_{j=S_i}^{\lambda} C_\lambda^j \rho^j (1 - \rho)^{\lambda-j}$. Therefore, the expectation of PLA is: $\sum_{j=S_i}^{\lambda} C_\lambda^j \rho^j (1 - \rho)^{\lambda-j} |w_i^m|$, where $|w_i^m|$ is the parameter amount of local model w_i^m for the m -th modality. Therefore, for the system contains N clients, the expectation of PLA can be denoted as: $\sum_{i=1}^N \sum_{j=S_i}^{\lambda} C_\lambda^j \rho^j (1 - \rho)^{\lambda-j} |w_i^m|$.

4.4 Attention-based Spontaneous Client Clustering

In the context of modality heterogeneity and device diversity, clients with identical modal configurations exhibit substantial data distribution heterogeneity, typically attributable to the use of different types of devices. This phenomenon is exemplified in Figure 10, which shows

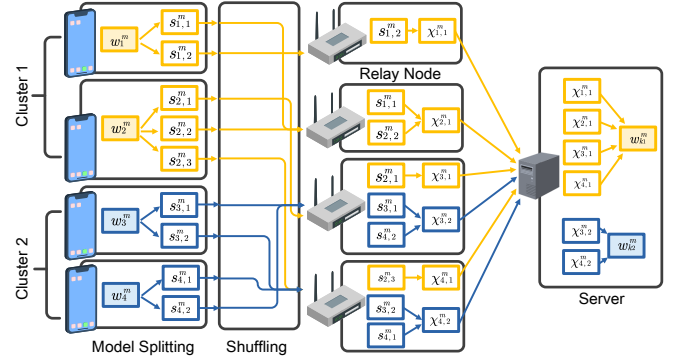


Figure 9: Workflow of split-shuffle mechanism.

the acceleration data distributions of four clients from the HHAR dataset [33]. We observe that clients using the same type of devices display more consistent data distributions. Interestingly, when aggregating the local model of a watch client (Client A) with phone clients and other watch clients, as shown in Figure 11, we observe a notable impact on the local test accuracy when models are aggregated across inconsistent data distributions. This insight underscores the importance of clustering clients with similar data distributions for more effective collaborative model learning. However, given the constraints of secure model aggregation where the cloud server cannot access client model plaintexts, traditional centralized clustering methods are impractical. To address this, we introduce Attention-based Spontaneous Client Clustering (ASCC), a novel distributed clustering scheme that maintains compatibility with the SSM and enables personalized clustering while preserving privacy.

The workflow of the ASCC is summarized in Algorithm 2. Specifically, after secure model aggregation, the cloud server broadcasts K cluster encoders $\{e_1^m, \dots, e_K^m\}$ for the modality m to each client n_i . To explore the fitness between local data distribution and K cluster encoders, each client n_i first uniformly samples a lightweight query set from its labeled dataset $\mathcal{D}_i^{m,1}$ with the proportion of $\beta \in (0, 1)$. The query set is denoted as $\mathcal{Q}_i^{m,0} = \{q_{i,1}^{m,0}, \dots, q_{i,\beta \cdot |\mathcal{D}_i^{m,1}|}^{m,0}\}$, where $q_{i,j}^{m,0}$ is the j -th query sample in $\mathcal{Q}_i^{m,0}$, and the superscript 0 means that $q_{i,j}^{m,0}$ is the original data feature. Via uniform sampling, $\mathcal{Q}_i^{m,0}$ is to approximate the true local data distribution of n_i .

Each client n_i randomly initializes KL_m trainable attention parameters, as the circles in Figure 12. L_m is the number of parameter layers of cluster encoders for the modality m . The attention parameters of n_i are denoted as $\mathcal{A}_i^m = \{A_i^{m,1}, \dots, A_i^{m,L_m}\}$, where $A_i^{m,l} = \{a_i^{m,l,1}, \dots, a_i^{m,l,K}\}$ represents the attention parameters of the l -th layer in K cluster encoders. Note that before attention training, all intrinsic parameters of cluster encoders are frozen to reduce computation and prevent the further fitting of cluster encoders to query data.

The forward calculation for each query sample $q_{i,j}^{m,0}$ is an iterative process. First, $q_{i,j}^{m,0}$ is input to the first

Algorithm 2: Attention-based Spontaneous Client Clustering

Input: (1) cluster encoders $\{e_1^m, \dots, e_K^m\}$ for the modality m ; (2) encoder layer number L_m for the modality m ; (3) query data sampling ratio β ; (4) labeled dataset $\mathcal{D}_i^{m,1}$; (5) learning rate μ

Output: local model e_i^m and c_i^m for the modality m

```

1 for  $n_i \in \mathcal{N}$  do
2   if  $m \in \mathcal{M}_i$  then
3     Freeze  $\{e_1^m, \dots, e_K^m\}$  and  $c_i^m$ ;
4     Initialize  $\mathcal{A}_i^m = \{A_i^{m,1}, \dots, A_i^{m,L_m}\}$ ;
5     Sample query data  $\mathcal{Q}_i^{m,0}$  from  $\mathcal{D}_i^{m,1}$  with  $\beta$ ;
6     /* Attention Training */
7     for  $(q_{i,j}^{m,0}, y_{i,j}^m) \in \mathcal{Q}_i^{m,0}$  do
8       for  $l = 1, \dots, L_m$  do
9         //normalize attention values
10         $\{a_i^{m,l,1}, \dots, a_i^{m,l,K}\} \leftarrow \text{Softmax}(A_i^{m,l})$ ;
11        Compute new query  $q_{i,j}^{m,l}$  with Eq. 9;
12         $q_{i,j}^{m,\text{out}} \leftarrow q_{i,j}^{m,L_m}$ ;
13        Compute attention loss  $L_{\text{att}}^j$  with Eq. 10;
14       $L_{\text{att}} \leftarrow \sum_{j=1}^{|\mathcal{Q}_i^{m,0}|} L_{\text{att}}^j$ ;
15      //update attention parameters
16       $\mathcal{A}_i^m \leftarrow \mathcal{A}_i^m - \mu \nabla L_{\text{att}}(\mathcal{A}_i^m)$ ;
17      /* Aggregation and Clustering */
18      Aggregate  $e_1^m, \dots, e_K^m$  with Eq. 12;
19      Re-select cluster according to Eq. 13;
20      /* Local Model Fine-tuning */
21      Unfreeze  $e_i^m$  and  $c_i^m$ ;
22      for  $(x_{i,j}^m, y_{i,j}^m) \in \mathcal{D}_i^{m,1}$  do
23        //compute classification loss
24         $L_{\text{cls}}^j \leftarrow F_{\text{CE}}(c_i^m(e_i^m(x_{i,j}^m)), y_{i,j}^m)$ ;
25       $L_{\text{cls}} \leftarrow \sum_{j=1}^{|\mathcal{D}_i^{m,1}|} L_{\text{cls}}^j$ ;
26      //update  $e_i^m$  and  $c_i^m$  with gradient descent
27       $e_i^m \leftarrow e_i^m - \mu \nabla L_{\text{cls}}(e_i^m)$ ;
28       $c_i^m \leftarrow c_i^m - \mu \nabla L_{\text{cls}}(c_i^m)$ ;
29 return local model  $e_i^m$  and  $c_i^m$  for the modality  $m$ 

```

layer of cluster encoders, i.e., $e_1^{m,1}$ and $e_2^{m,1}$ in Figure 12. Then the intermediate features are calculated as $f_{i,j}^{m,1,k} = e_k^{m,1}(q_{i,j}^{m,0})$. After that, the attention parameters $A_i^{m,1} = \{a_i^{m,1,1}, \dots, a_i^{m,1,K}\}$ for the first layer of K cluster encoders are normalized by softmax, and then the intermediate features $\{f_{i,j}^{m,1,1}, \dots, f_{i,j}^{m,1,K}\}$ are weighted by A_i^1 to obtain the new query $q_{i,j}^{m,1} = \sum_{k=1}^K a_i^{m,1,k} f_{i,j}^{m,1,k}$. The superscript 1 in $q_{i,j}^{m,1}$ represents the feature calculated by the 1st parameter layer. In this way, the iterative equation on $q_{i,j}^{m,l}$ is obtained as:

$$q_{i,j}^{m,l} = \sum_{k=1}^K a_i^{m,l,k} \cdot e_k^{m,l}(q_{i,j}^{m,l-1}), a_i^{m,l,k} \in A_i^{m,l}. \quad (9)$$

The final output is $q_{i,j}^{m,\text{out}} = q_{i,j}^{m,L_m}$, and the attention loss of the j -th query sample is calculated as

$$L_{\text{att}}^j = F_{\text{CE}}(c_i^m(q_{i,j}^{m,\text{out}}), y_{i,j}^m). \quad (10)$$

where F_{CE} is the cross entropy loss function. After forward processing, n_i applies the Stochastic Gradient Descent

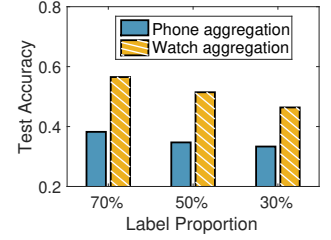
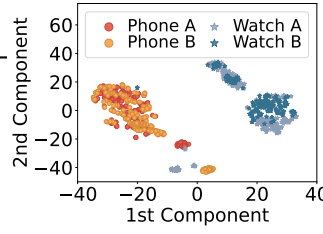


Figure 10: Data distribution of clients using phones and watch client A applying different aggregation strategies.

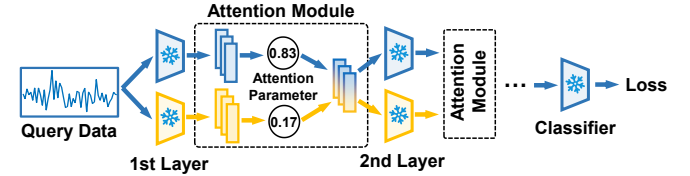


Figure 12: Workflow of attention training.

(SGD) to minimize the attention loss and update the attention parameters $\mathcal{A}_i^m$ till convergence by:

$$\mathcal{A}_i^{m*} = \arg \min_{A_i^{m,1} \dots A_i^{m,L_m}} \sum_{j=1}^{|\mathcal{Q}_i^{m,0}|} L_{\text{att}}^j. \quad (11)$$

After attention training, n_i takes the updated attention parameters $\mathcal{A}_i^{m*}$ as the aggregation weights of K cluster encoders in the m -th modality, and then gets the personalized local encoder e_i^m :

$$e_i^m = \bigoplus_{l=1}^{L_m} \sum_{k=1}^K a_i^{m,l,k} \cdot e_k^{m,l}. \quad (12)$$

where $\bigoplus$ is defined to concat aggregated parameter layers.

Attention parameters indicate the fitness between local data distribution and each cluster. In other words, clients are more likely to find similar collaborator in the cluster with higher fitness. To re-select better cluster, n_i sums all attention parameters of each cluster model to an accumulated attention value, and select the cluster with highest one, as:

$$k_{\text{max}}^i = \arg \max_k \left\{ \sum_{l=1}^{L_m} a_i^{m,l,k} \mid a_i^{m,l,k} \in A_i^{m,l}, k \in [1, K] \right\}. \quad (13)$$

After getting the personalized encoder e_i^m , each client n_i uses the labeled data $\mathcal{D}_i^{m,1}$ to fine-tune the local model, including e_i^m and c_i^m .

In MASA, the ASCC enhances privacy, spontaneity in clustering, layer-wise personalization, and maintains computational efficiency. ASCC ensures stringent privacy preservation by not requiring clients to disclose their model plaintexts or data distributions, using cluster models that amalgamate information from multiple clients without compromising individual privacy. Unlike traditional server-dependent clustering methods, ASCC allows clients to self-organize into clusters through local training and adaptive attention, suitable for environments without a trusted server. It introduces layer-wise personalization by allowing varying aggregation weights for different parameter layers, aligning with the understanding that data heterogeneity

Table 2: Summary of Selected Multimodal Datasets

Dataset	Class	Subject	Modality	Sample
HHAR	6	9	Acc, Gyro	2052
MHAD	11	12	IMU, Skeleton	3600
MHealth	12	10	Acc, Gyro	6812
mmFi	27	40	Depth, Skeleton, mmWave	15120

impacts layers differently, especially deeper ones more sensitive to data distribution. Moreover, thanks to the freeze of cluster model parameters during attention training, the backward propagation of gradients can be simplified technically, and the computational cost of optimizer update is also reduced.

5 EXPERIMENT ANALYSIS

In this section, we present the detailed evaluation methodology and results.

5.1 Evaluation Methodology

Datasets. We select four representative public multimodal datasets of human activity recognition tasks involving multiple sensors and subjects, and summarize them in Table 2. The details of each dataset are shown as follows.

HHAR [33]. The HHAR dataset features data from 9 participants who used a variety of smartphones (such as Nexus, Samsung Gold, etc.) and smartwatches (including LG Watch, Gear, etc.) to record acceleration and gyroscope data across 6 categories of activities. Each sample sequence in the dataset has been standardized to a length of 50, yielding a total of 2052 samples.

MHAD [6]. The MHAD dataset comprises both IMU and skeleton data sequences, capturing 12 subjects engaged in 11 different activity classes. The IMU data encompasses 9 axes, and the skeleton data precisely records the 3-dimensional coordinates of 35 human body joints. To standardize our analysis, we adjusted the length of each sample sequence to 60, culminating in a comprehensive collection of 3600 samples.

MHealth [41]. The MHealth dataset captures data from 12 distinct activity classes performed by 10 subjects. Throughout the data collection phase, three sensors were strategically placed on each subject—at the chest, left ankle, and right arm—to collect both 3-axis acceleration and 3-axis gyroscope data. MHealth meticulously records the original time series for each activity undertaken by the subjects. For analysis, we standardized the sample sequence length to 50, enabling us to extract a total of 6812 samples from the original sequences.

mmFi [39]. The mmFi dataset amasses activity data across 27 classes from 40 subjects, spanning 5 modalities: RGB, depth image, skeleton, lidar, and mmwave. We specifically focus on three modalities—depth image, skeleton, and mmwave—to construct client datasets with diverse modality combinations. Thanks to meticulous synchronization during multimodal data collection, the length of sample sequences is uniformly set to 30 across all modalities, resulting in a total of 15120 samples. To prevent overfitting and enhance convolutional feature extraction, each depth

Table 3: Comparison of MASA with baseline unimodal FL approaches

Dataset	HHAR	MHAD	MHealth	mmFi
FedAvg	34.9±1.5	70.7±1.0	58.3±2.2	69.6±2.6
FedProx	36.9±2.6	69.3±3.3	53.8±2.0	71.7±3.8
FedPer	39.3±2.5	52.1±2.7	75.2±2.2	60.5±1.5
LG-Fed	50.4±2.2	68.3±2.1	62.4±2.1	73.4±1.4
CFL	52.0±2.2	56.9±2.2	69.3±2.8	71.6±1.8
MASA (Ours)	55.6±1.3	72.9±2.9	80.6±1.6	83.8±1.0

image frame is resized to 80×60 pixels. Furthermore, following [39], we preprocess each mmwave sequence frame into 5 feature maps of 8×8 dimensions.

Baselines. We compare MASA against seven baseline FL frameworks, which include FedAvg [46] for aggregating client models based on data volume, FedProx [17] that introduces regularization to align local and global models, and FedPer [1], which separates feature extractors and classifiers for personalization. LG-Fed [19] contrasts with FedPer by aggregating classifiers. CFL [5] forms clusters of clients according to model similarities. Additionally, MultiFL [44] and Harmony [25] address multimodal scenarios, with the former supporting modality fusion and the latter combining modality-specific training with personalized tuning. These baselines provide a comprehensive assessment framework, covering various aspects of federated learning such as model aggregation, personalization, and multimodal data handling.

Implementation. The MASA and baseline models are implemented in PyTorch and trained on a server equipped with an NVIDIA GTX 3090 GPU. For feature extraction, we utilize 1D CNN layers for IMU, accelerometer, and gyroscope data; 2D CNN layers for skeleton data; and 3D CNN layers for depth images and mmWave data, respectively. During the local training phase, the batch size is set to 16. The learning rate is initially set at 0.001 for depth data and 0.01 for other modalities, with a linear decay applied throughout the system rounds. Furthermore, in the four datasets we have selected, each client's original record for each action is represented as a temporal sequence. We utilize a sliding window of 50 frames to extract multiple consecutive subsequences of the same length from these sequences, using them as data points for the respective actions. It is important to note that for clients with multimodal data, the sampling windows are aligned across different modalities. Following the extraction process, data points for each client are normalized individually. The data points are then randomly divided into a training set and a test set in a 7:3 ratio. Additionally, the training set is further randomly split into a labeled set and an unlabeled set, with the default ratio being 50%. During the training phase, the labels of all samples in the unlabeled set are made invisible to the clients. To ensure the reliability of our results, each experiment is conducted five times.

5.2 Overall Performance

We compare MASA with two leading multimodal FL methods, specifically MultiFL and Harmony. To ensure an equitable comparison, we maintain consistent training and testing configurations for these systems in line with MASA.

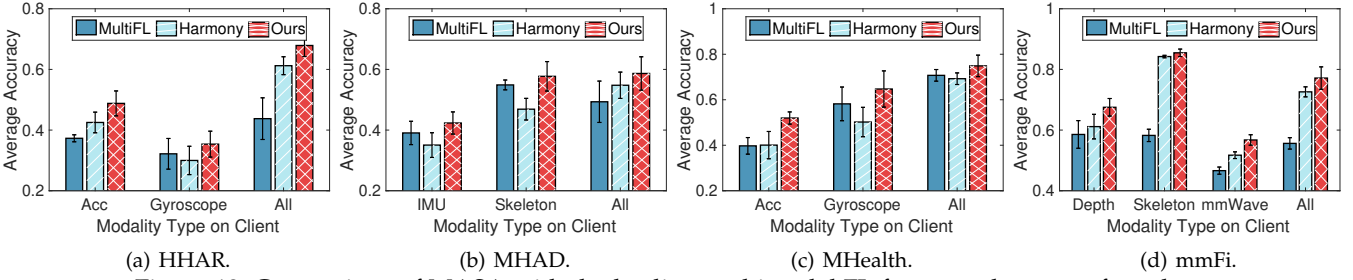


Figure 13: Comparison of MASA with the leading multimodal FL frameworks across four datasets.

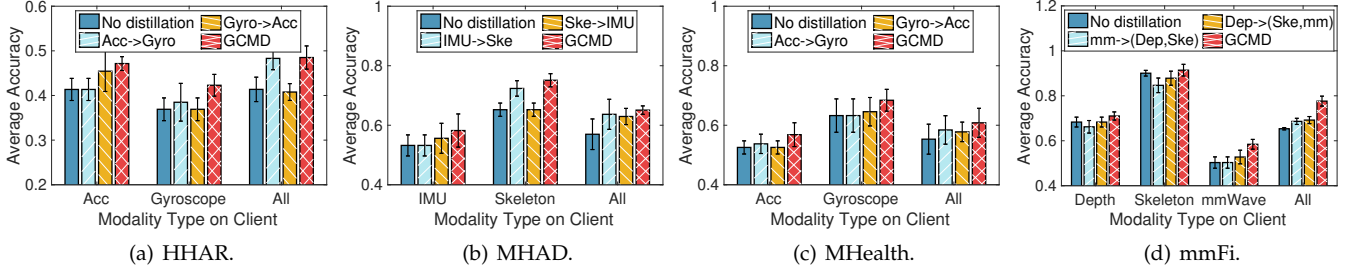


Figure 14: Comparison of MASA with no distillation or fixed distillation across four datasets.

The experimental outcomes are presented in Figure 13, illustrating the average classification accuracy across clients with varying modal configurations (i.e., clients with heterogeneous modalities). We note that MultiFL typically underperforms compared to the other methods, which is attributed to the integration of fused multimodal and unimodal models, which results in damaging gradient inconsistencies. Harmony seeks to address this challenge using a late fusion mechanism, yet it fails to adequately consider the performance of unimodal clients and the diversity in data distributions. In clients with assorted modal setups, MASA markedly surpasses the other techniques. MASA exhibits an average accuracy improvement across four datasets over MultiFL and Harmony by 11.16% and 8.89%, respectively. This enhancement predominantly stems from our two central designs: GCMD and ASCC. Particularly during the local training stage, MASA ensures effective modality interaction among multimodal clients via GCMD, thereby boosting inferior single-modality models with the help of superior counterparts. Additionally, MASA employs ASCC in the global parameter aggregation phase to counter the adverse effects of data distribution heterogeneity on the fusion of models. In summary, the results validate the efficacy of our proposed multimodal FL approach in environments characterized by modal heterogeneity.

Moreover, Table 3 shows the comparison of our MASA and unimodal FL baselines on the final average testing accuracy. Among them, FedAvg and FedProx both uniformly aggregates all client models to a single global model, which is easy to implement but ignores the interaction among modalities and potential data distribution heterogeneity of clients. As two classical approaches of layer decoupling, FedPer and LG-Fed retain partial local parameters to reach the balance of generalization of personalization. However, these two approaches easily encounter the problem of overfitting, especially when the labeled data is insufficient. CFL allows clients with similar data distributions to aggregate their local models, thus conflicting training objectives. How-

ever, CFL requires the data distribution information or model parameters of all clients as clustering prior, which to some extent violates the privacy preservation principle in FL. In addition, the models trained on different modalities are still completely isolated, which hinders the potential improvement brought by proper modality interaction.

5.3 Ablation Study

We conduct two ablation studies to validate the effectiveness of our key components, i.e., GCMD and ASCC.

5.3.1 Performance of GCMD

Figure 14 illustrates the average model accuracy of unimodal and multimodal clients applying various distillation strategies. Typically, compared to no distillation and fixed distillation, GCMD demonstrates superior model performance in multimodal clients. This is attributed to GCMD's ability to evaluate and adaptively select the most suitable distillation direction based on the superiority of modalities. For example, in the mmFi dataset (as shown in Figure 14 (d)), the skeleton and depth image modalities are clearly more effective than mmwave. GCMD recognizes and leverages this advantage. Consequently, under GCMD, the superior knowledge from skeleton and depth image modalities is effectively transferred to enhance mmwave models. In contrast, if the distillation direction is preset from mmwave to other modalities, the skeleton and depth models do not achieve significant improvement, due to the limited utility of knowledge from mmwave. The improvement in model accuracy with GCMD is 12.3%, whereas a fixed distillation strategy yields only a 3.3% or 3.9% improvement over no distillation.

Furthermore, for unimodal clients, especially those with inferior modalities, their model performance also benefits from model aggregation with enhanced models from multimodal clients via GCMD. For instance, acceleration clients in the HHAR dataset and mmwave clients in the mmFi dataset experience improvements of 6.2% and 8.1% in

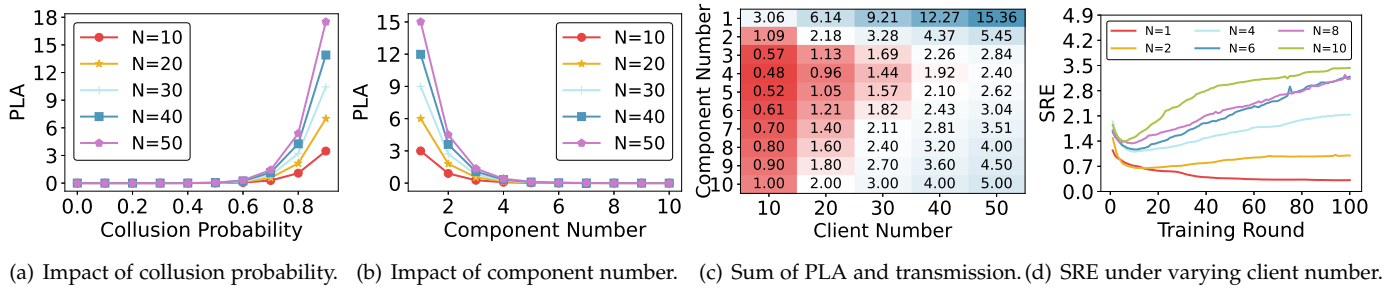


Figure 15: Overall defense performance of SSM.

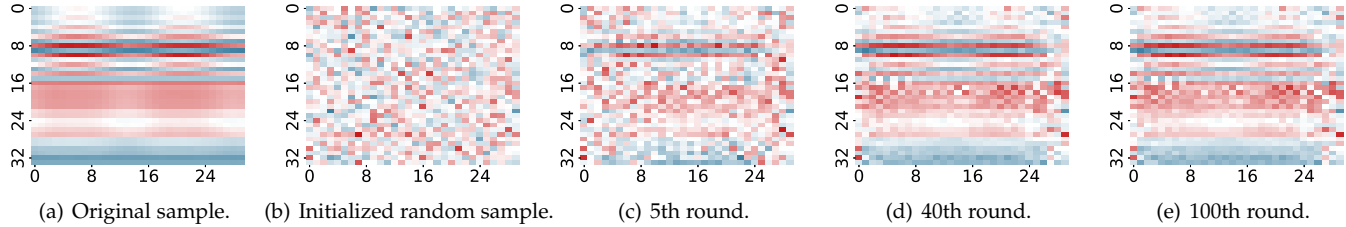


Figure 16: Sample reconstruction process of DGL attack on the skeleton modality in mmFi dataset.

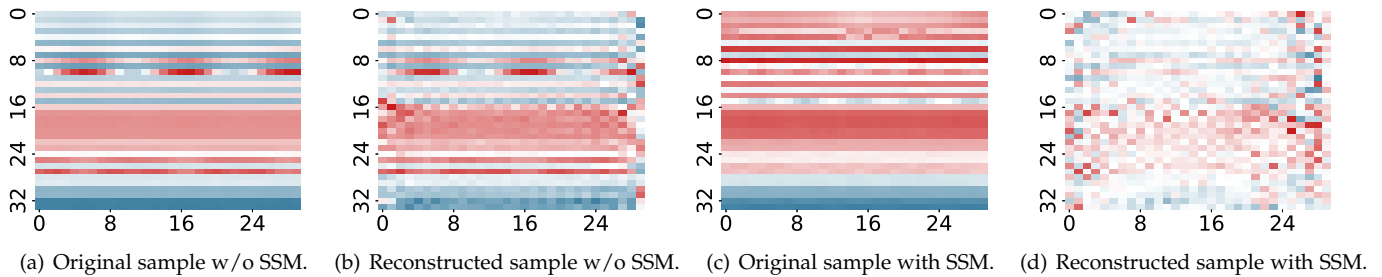


Figure 17: Comparison of sample reconstruction effect with (without) SSM.

model accuracy, respectively, compared to scenarios without distillation. For unimodal clients with inferior modalities, this is akin to effectively utilizing superior modalities from other clients with heterogeneous modality configurations. Overall, our proposed GCMD strategy not only optimizes the local training of multimodal clients but also indirectly enhances the model performance of unimodal clients.

5.3.2 Performance of SSM

We first analyze the impact of different configurations in SSM on client privacy preservation. Figure 15(a) and Figure 15(b) show the total number of client model parameters recovered by the server and collusive relay nodes, i.e., the PLA indicator defined in the attack model, under varying collusion probability and random component number. We repeated the SSM process for times and calculated the mean value of PLA. From Figure 15(a), when the random component number is fixed, PLA constantly increases with the collusion probability that represents the strength of the collusive attack. In other words, the larger collusion probability, the more relay nodes collude with the server, and the higher the probability that all the selected relay nodes by the clients are untrusted. Secondly, when the collusion probability is fixed, PLA continuously decreases as the random component number increases, which indicates that splitting the local model into more random components for distributed storage is more secure. However, when the component number is too large, the transmission load from the client

to the relay nodes will be more severe. Figure 15(c) shows the weighted sum of transmission load and PLA under varying component number and client number, from which we conclude that choosing an appropriate contributes to the balance of privacy security and communication efficiency. Then we analyze the defense performance of SSM against DGL attacks. Figure 15(d) compares the error between the samples recovered by the server and the corresponding original samples, i.e., the SRE indicator defined in the attack model. The client number in Figure 15(d) represents the number of clients participating in the cluster model aggregation. Specifically, client number=1 indicates that the server can obtain the local model of clients, and the private samples can be well reconstructed via the DGL attack, shown as Figure 16. Generally, SRE increases with the client number in Figure 15(d), indicating that the cluster models effectively perturb individual client data patterns to prevent the server from recovering samples from any client. This echoes the goal of SSM that the server is able to access only the cluster models rather than the local models of clients. Figure 17 compares the samples recovered by the server applying and not applying SSM. We observe that if the server obtains the local model of clients, i.e., without SSM, the patterns of training samples can be basically recovered. For cases applying SSM, that is, the server cannot obtain the local models of individual clients, there is more noise in the reconstructed samples, and even the important patterns are lost. Overall, SSM enables secure aggregation of client

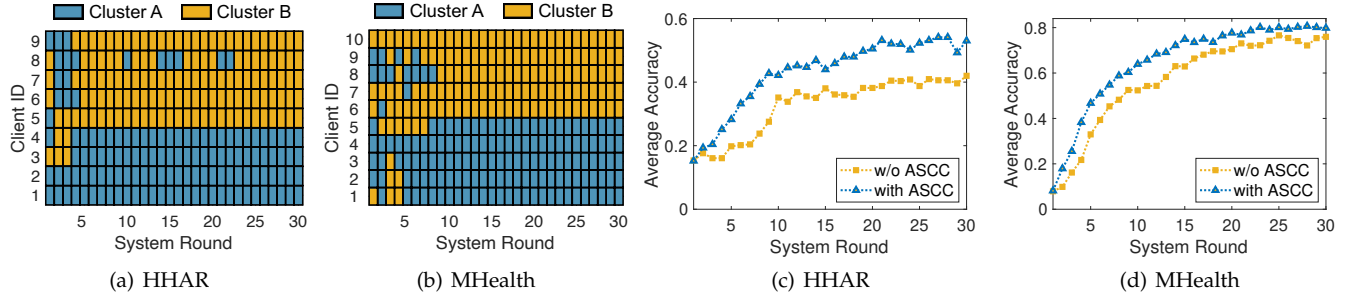


Figure 18: Performance of the ASCC, where (a) (b) represents the clustering results, (c) (d) represents the system convergence.

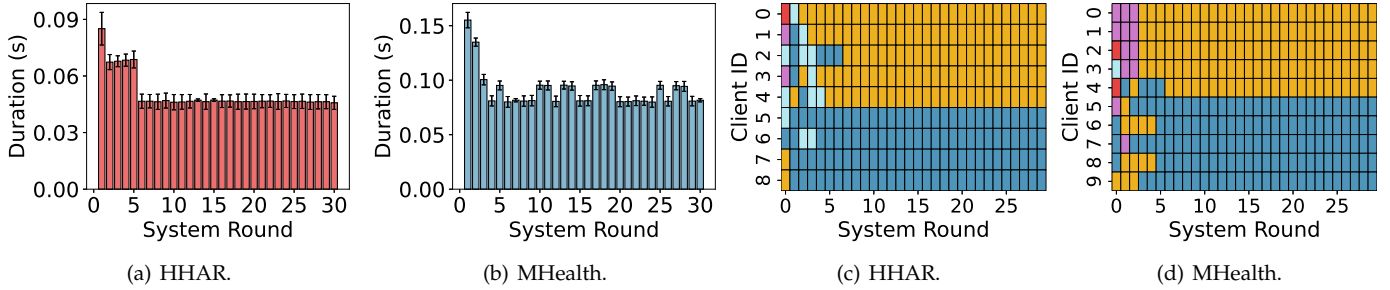


Figure 19: Performance of ASCC with large initialized cluster number, where (a) (b) represents the computational costs, (c) (d) represents the clustering results.

models on the server without disclosing any individual client models, which is effective against privacy attacks.

5.3.3 Performance of ASCC

In this study, we select two representative datasets, MHealth and HHAR, characterized by significant data distribution heterogeneity. This heterogeneity, primarily due to the varying user wear habits of accelerometers and gyroscopes, presents an ideal testbed for our proposed ASCC method. Our method aims to achieve secure and spontaneous client clustering based on distinct data distributions, as illustrated in Figure 18 (a) and Figure 18 (b). At the outset of the system initialization, clients, devoid of prior knowledge about their counterparts' data distribution, are randomly assigned to clusters, with the cluster affiliation denoted by the block color of each client. ASCC enables a dynamic clustering mechanism. Through iterative cycles of local training and aggregation within clusters, it systematically guides clients with similar data distributions to form and stabilize in cohesive clusters. Once established, these clusters maintain stability throughout subsequent rounds of the system's operation. For instance, within the MHealth dataset, clients having sensors affixed in similar positions naturally cluster together, thereby aggregating their respective models. Similarly, in the HHAR dataset, clients using the same type of sensors collaboratively fine-tune a robust prediction model, ensuring uniformity in data distribution and training objectives. A critical aspect of the client clustering formation is the confidentiality of model parameters. At no point are these parameters shared in plaintext, ensuring robust protection of client privacy. Moreover, Figure 18 (c) and Figure 18 (d) depict the comparative analysis of the average accuracy of client models with and without the implementation of ASCC in both MHealth and HHAR datasets. Unlike the conventional approach of aggregating a single global model for clients sharing the same modality,

our ASCC method introduces distribution-aware client clustering and layered model personalization. This approach not only accelerates the convergence in federated training but also significantly enhances the peak accuracy of client models.

5.4 Computational efficiency of ASCC

Additionally, our analysis explores the time-cost expenses of ASCC across various training phases, including unsupervised feature extraction, supervised fine-tuning, and attention training. This comparative review is systematically detailed in Table 4. Delving deeper, we utilize the mhealth and mmFi datasets to track the cumulative computational times of different client modalities, measured in seconds. These times are meticulously recorded across two different CPU configurations: AMD Ryzen 5 4600H and Intel Xeon Platinum 8222CL. The findings reveal significant differences in computational demands across the learning stages; notably, the computational time required for the attention training phase is significantly reduced, accounting for only 1.77% to 5.96% of the total local computation time.

Then we analyze the impact of initialized cluster number on the computational cost of client devices in attention training. Figures 19(a)-(b) respectively show the changes of average attention training duration in each system round for the MHealth dataset and HHAR dataset. We observe that in the initial stage of iterative spontaneous clustering, due to the large initialized cluster number, each client needs to handle more cluster models in parallel, resulting in a larger attention training duration. However, in the subsequent stage, the attention training duration shows a downward trend. To investigate the reason, we visualized the cluster labels of clients during the clustering when the initialized cluster number is 5, shown as Figures 19(c)-(d). We found that the initialized cluster number automatically decreases from 5 to 2 in the iterative clustering. This is because, accord-

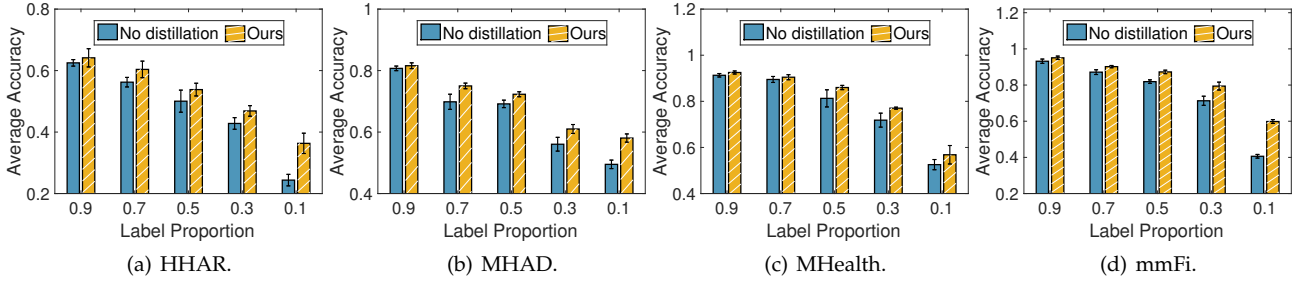


Figure 20: Performance of MASA with or without cross-modal knowledge distillation at various label proportions across four datasets.

Table 4: Computation time of clients with different modality combinations

Dataset	MHealth			mmFi			
	Acc	Gyro	All	Depth	Skeleton	mmWave	All
Unsupervised (s)	42.38	40.62	77.31	91.27	7.71	4.86	107.52
Supervised (s)	4.28	4.50	7.19	21.07	5.28	3.38	29.48
Attention (s)	1.17	0.94	1.52	7.92	0.88	0.74	8.69
Total (s)	47.83	46.06	86.02	120.26	13.87	8.98	145.69
Attention training cost proportion (%)	2.45%	2.04%	1.77%	6.59%	6.34%	8.24%	5.96%

ing to our experimental setup, clients in both the MHealth dataset and HHAR dataset can be divided into two heterogeneous subgroups with explicit characteristics. In attention training and cluster re-selection, each client independently selects the cluster model with the highest attention value and join in the corresponding cluster patterns. If no client chooses a certain cluster, the corresponding cluster model will be naturally discarded and the cluster number will automatically decreases. In this way, the number of cluster models need to be handled on client devices is reduced, thereby alleviating the computational load on client devices.

The above results highlight the reduced computational burden offered by attention training. The efficiency improvement is primarily due to the optimized query set sampling process and the moderate number of attention parameters needed, thereby confirming the role of attention training as a cost-effective element in the local training framework.

5.5 Impact of Label Proportion

In our analysis of MASA's performance across various label proportions, as depicted in Figure 20, we observe notable trends. Specifically, with a reduction in the supervision signal, modalities with inherently lower performance struggle to develop high-quality models relying solely on limited labeled data. For instance, in the absence of cross-modality distillation, the mmWave modality in the mmFi dataset achieves a mere 25.6% testing accuracy under a label proportion of 0.1. This scenario highlights a significant performance discrepancy compared to the other two superior modalities. Conversely, MASA adeptly leverages these performance disparities in scenarios with scarce labeled data, enabling superior modalities to aid in enhancing the performance of their inferior counterparts. For example, under a label proportion of 0.1, the mmWave model, when subjected to cross-modal distillation, shows an improvement of 19.2% in accuracy compared to training in isolation. In conclusion, our methodology demonstrates a pronounced performance

enhancement in environments with limited labeled data, rendering it particularly effective for multimodal sensing applications.

6 DISCUSSION

6.1 Limitations of Cross-Modal Distillation

The current cross-modal knowledge distillation technique in MASA utilizes discrepancies in performance among different modalities during unsupervised learning to enhance the less effective modality. In such setups, for multimodal configurations with minimal performance differences, like Acc and Gyro in MHealth, these modalities alternately act as the superior modality to aid the other. Consequently, the application of GCMD effectively uses both modalities to enhance each other's performance. However, for configurations with a substantial performance disparity, such as Skeleton and mmWave in mmFi, Skeleton consistently outperforms mmWave. Under the control of the distillation gate, Skeleton has limited opportunities to gain effective knowledge from mmWave. Forcing substandard knowledge from mmWave onto Skeleton could even degrade Skeleton's performance below the baseline in the absence of distillation. Therefore, in such scenarios, GCMD is beneficial only to the poorer modality. A potential solution to this limitation could be the development of more efficient cross-modal knowledge distillation techniques, which forms a part of our future research agenda.

6.2 Computational Efficiency

Furthermore, the incorporation of additional security measures invariably influences the performance of other facets, particularly computational efficiency. While the implementation of straightforward encryption methods and low-cost attention training ensures minimal computational burden in the ASCC framework, it nonetheless necessitates a trade-off in terms of communication. This trade-off arises from the necessity to split local models and broadcast multiple

cluster models, which can become problematic especially when dealing with large model sizes or limited bandwidth resources. Such circumstances may lead to increased communication overhead, subsequently impairing the efficiency of model aggregation. Therefore, in light of ASCC's communication constraints, employing model quantization or compression techniques may provide a more balanced approach, enhancing computation, communication, and privacy security within FL systems. This approach not only aims to optimize resource use but also seeks to uphold the stringent requirements of data privacy and model integrity.

7 CONCLUSION

This paper presents MASA, a multimodal sensing FL framework that enhances the model performance of both multimodal and unimodal clients while ensuring their data privacy. To achieve this objective, MASA incorporates three advanced techniques. First, a gated cross-modal distillation scheme is deployed at multimodal clients to leverage the knowledge from superior modalities to boost inferior ones, which can be further transferred to unimodal clients via model aggregation. To resist privacy attacks, MASA integrates a split-shuffle mechanism to achieve secure model sharing. Compatible with split-shuffle mechanism, an attention-based spontaneous client clustering mechanism is designed to construct client cluster structures without disclosing client model plaintexts to promote personalized collaboration. The extensive experiments demonstrate that our MASA has excellent potential for multimodal sensing scenarios such as human activity recognition.



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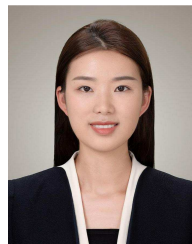
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